matsapphouse scripts

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Script 1: "Nine Brains? (Octopus Arms)"

Format: 45–60s vertical short + 30s song. **Topic:** neuroscience

Hook (0–3s)

"Two-thirds of an octopus's neurons aren't in its head."

Shot list

1. Close-up of an octopus eye in a tank (or a stock clip/animation). Text: "~500 million neurons".
2. Pie chart animation: about 2/3 in the arms, 1/3 in the central brain.
3. One arm grabbing a shrimp while the head looks elsewhere.
4. Diagram of a nerve cord running down each arm, with ganglia at the suckers.
5. Cut-arm research card (narrated, no gore): a severed arm still reaches and grasps for a while.
6. Map of the arm cord split into segments (2022 study).
7. Mat to camera: "So does the arm think? It handles the details. The brain sets the goal."
8. Song outro over tentacle loops.

Lyrics: "Eight Arms, One Plan"

Verse Half a billion neurons, most aren't in my head, Eight long arms that taste the sea and work things out instead. I say "find a crab," and down below Every sucker reads the rock and knows which way to go.

Chorus Eight arms, one plan, The brain says "what," the arms say "how," Eight arms, one plan, Thinking in the reef right now.

Verse 2 A cord in every arm, with knots at every cup, They smell and grab and pass it on and never give it up. Cut the line and an arm can still reach and close its hand. It isn't quite a mind, but it's more than you had planned.

Science notes

- About 500M neurons, around two-thirds in the arms: Hochner, *Curr Biol* 2012, "An embodied view of octopus neurobiology". https://doi.org/10.1016/j.cub.2012.09.001
- Severed arms still reach and grasp: Sumbre et al., "Control of octopus arm extension by a peripheral motor program", *Science* 293:1845 (2001). https://doi.org/10.1126/science.1060976
- Segmented axial nerve cord: Olson et al., "Neuronal segmentation in cephalopod arms", *Nature Communications* 16:443 (2025; preprint bioRxiv 2024). https://doi.org/10.1038/s41467-024-55475-5
- Suckers have chemotactile receptors: van Giesen et al., *Cell* 2020. https://doi.org/10.1016/j.cell.2020.09.008
- Careful with wording: "nine brains" is a popular simplification. Say "partly independent arms," not "separate brains."

Script 2: "Why Do We Dream?"

Format: 60s short + song. **Topic:** neuroscience

Hook

"Every night your body is paralysed so you can't act out your dreams."

Shot list

1. Bedroom at night with a phone clock reading 3:00. Text: "REM sleep".
2. Sleep-cycle graph: about 90-minute cycles, with REM getting longer toward morning.
3. Brain cutaway: the visual and emotional areas (amygdala) light up, and the prefrontal area dims (which is why dreams feel strange).
4. Brainstem "off switch" animation: muscle atonia.
5. Three theory cards: memory consolidation; emotional processing ("overnight therapy"); threat rehearsal. The caption says "still debated".
6. Rat-maze replay clip (animation): hippocampal replay.
7. Mat: "Short answer: nobody's fully sure, but sleep helps memory a lot."

Lyrics: "Night Shift"

Verse Lights go out at the edge of the day, My brain clocks in while my body stays, The dimmer's down on the front of my mind, So a fish can drive and it seems just fine.

Chorus Night shift, filing the day, Replaying the maze till it's here to stay, Night shift, soften the fear, Wake up lighter, the morning's here.

Bridge Ninety minutes, round we go, REM gets longer as the hours go.

Science notes

- REM atonia and ~90-min cycles: Peever & Fuller, *Curr Biol* 2017. https://doi.org/10.1016/j.cub.2017.10.026
- Prefrontal deactivation and limbic activation in REM: Maquet et al., *Nature* 1996. https://doi.org/10.1038/383163a0
- Hippocampal replay in rats: Wilson & McNaughton, *Science* 1994. https://doi.org/10.1126/science.8036517
- "Overnight therapy": Walker & van der Helm, *Psychol Bull* 2009. https://doi.org/10.1037/a0016570
- Threat simulation theory: Revonsuo, *Behav Brain Sci* 2000.
- Present the purpose of dreaming as open, not settled.

Script 3: "The Mantis Shrimp Myth"

Format: 45s short + song. **Topic:** animal kingdom

Hook

"Mantis shrimp have 12 to 16 colour receptors, but they're worse at telling colours apart than you are."

Shot list

1. Mantis shrimp clip with rainbow text "12+ photoreceptors!"
2. Record scratch. Text: "...but".
3. Comparison graphic: human 3 cones vs. mantis 12. The mantis fails to tell apart colours about 15–25 nm apart (Thoen 2014).
4. Theory card: fast "barcode" colour reading instead of comparing colours.
5. Punch strike in slow motion: the club accelerates like a .22 bullet and makes cavitation bubbles.
6. Mat: "Not super-vision. Super-fast vision."

Lyrics: "Barcode Eyes"

Verse Twelve kinds of colour in a candy-striped eye, You'd think I see a rainbow you could never buy, But put two blues together and I can't tell, I'm reading the world like a checkout bell.

Chorus Barcode eyes, beep beep beep, No time to think in the coral deep, Barcode eyes, fast not fine, One quick scan and the reef is mine.

Verse 2 And when I swing, the water boils, Bubbles pop, the shell recoils.

Science notes

- Poor colour discrimination despite 12 receptors: Thoen et al., *Science* 2014. https://doi.org/10.1126/science.1245824
- Strike mechanics and cavitation: Patek et al., *Nature* 2004. https://doi.org/10.1038/428819a ; Patek & Caldwell, *J Exp Biol* 208:3655 (2005). https://doi.org/10.1242/jeb.01831
- Some species have 16 photoreceptor types in total (12 for colour, the rest for polarisation and UV): Cronin & Marshall, "Parallel processing and image analysis in the eyes of mantis shrimps", *Biol Bull* 200:177 (2001). https://doi.org/10.2307/1543312

Script 4: "A Teaspoon of Neutron Star"

Format: 45s short + song. **Topic:** astrophysics

Hook

"A teaspoon of this star would weigh about as much as a mountain."

Shot list

1. Teaspoon on a kitchen counter, with a glowing animated blob in it.
2. Scale card: the Sun's mass squeezed to the size of Osaka (about 20 km across).
3. Collapse animation: a supernova; protons and electrons merge into neutrons.
4. Spin card: the fastest known pulsar spins 716 times per second (PSR J1748-2446ad).
5. Beam "lighthouse" animation with a pulsar sound clip.
6. Gold card: the 2017 neutron star merger GW170817 made heavy elements like gold.
7. Mat holding a ring: "This may have come from a neutron star smash."

Lyrics: "Teaspoon Star"

Verse A sun crushed down to a city wide, No room for atoms, nowhere to hide, Electrons fall and protons too, Now it's neutrons all the way through.

Chorus Teaspoon star, heavy as a hill, Spinning so fast and never still, Seven hundred times a second, round, Lighthouse beams without a sound.

Bridge Two collide and the ripples spread, Gold rains out, they're sending us our wedding thread.

Science notes

- Density ~10^17 kg/m³. A 5 mL teaspoon ≈ 5×10^11 kg, which is in the range of a large mountain (estimates vary). NASA: https://imagine.gsfc.nasa.gov/science/objects/neutron_stars1.html
- Radius ~10 km, mass ~1.4 Suns: same NASA page.
- 716 Hz pulsar: Hessels et al., *Science* 2006. https://doi.org/10.1126/science.1123430
- GW170817 kilonova and r-process elements: Abbott et al., *ApJL* 2017. https://doi.org/10.3847/2041-8213/aa91c9 . Kilonovae are one source of gold; how much they contribute is still debated.

Script 5: "Photo of the Dark"

Format: 50s short + song. **Topic:** astrophysics

Hook

"This is a real photo of something that light can't escape."

Shot list

1. Black screen, then the orange ring of M87* fades in.
2. Scale card: M87* is about 6.5 billion times the Sun's mass, about 55 million light-years away.
3. Globe animation: radio dishes from Hawaii to the South Pole linking up into one Earth-sized telescope.
4. Mat with a hard drive: "Too much data for the internet, so they flew the hard drives."
5. Shadow explainer: the dark centre is the black hole's shadow; the ring is hot gas bent by gravity.
6. 2022 card: our own galaxy's black hole, Sagittarius A*, about 4 million Suns, about 27,000 light-years away.
7. Side-by-side rings: "Same physics, a thousand times different in size."

Lyrics: "Photo of the Dark"

Verse Dishes on mountains, ice and sea, Turned the whole Earth into one big eye to see, Petabytes flown on a plane, Stitched it together, again and again.

Chorus Photo of the dark, a ring of fire, Light bent round and pulled up higher, Nothing gets out, but we caught its shape, A shadow no light can escape.

Bridge Home in the Milky Way, one more ring, Einstein smiling at everything.

Science notes

- M87 *shadow image, mass ~6.5×10^9 Suns, distance ~16.8 Mpc (~55 million ly), 1.3 mm observations: Event Horizon Telescope Collaboration,* ApJL* 875:L1 (2019). https://doi.org/10.3847/2041-8213/ab0ec7
- Sgr A *image, mass ~4×10^6 Suns, distance ~8 kpc (~27,000 ly): Event Horizon Telescope Collaboration,* ApJL* 930:L12 (2022). https://doi.org/10.3847/2041-8213/ac6674
- Data were shipped on hard drives rather than sent over the internet (the 2017 run recorded petabytes): described in EHT 2019 Paper I and EHT press materials. https://eventhorizontelescope.org
- The ring is light from hot gas bent around the black hole; the dark centre is the shadow, not the event horizon itself: EHT 2019 Paper I.

Script 6: "Inner GPS"

Format: 50s short + song. **Topic:** neuroscience

Hook

"Your brain has a map of this room, and it's drawn in hexagons."

Shot list

1. Mat walks across a room; a glowing dot follows him on an overhead floor plan.
2. Card: 1971, John O'Keefe records single neurons in a rat's hippocampus.
3. Animation: one "place cell" fires only when the rat stands in one spot.
4. Card: 2005, May-Britt and Edvard Moser find "grid cells" next door in the entorhinal cortex.
5. Overlay: each grid cell's firing spots form a triangular grid of hexagons tiling the floor.
6. Nobel medallion: 2014 Nobel Prize in Physiology or Medicine for all three.
7. Mat taps his head: "No satellites needed."

Lyrics: "Inner GPS"

Verse Step to the window, one cell sings, Step to the door, a new one rings, Place by place, the map gets drawn, Even in the dark, it carries on.

Chorus Inner GPS, hexagon floor, Grid cells counting every step and more, I know where I am, I know where I've been, A map of the world that lives within.

Bridge Three scientists, one prize in gold, The oldest map that's never been sold.

Science notes

- Place cells: hippocampal neurons firing when a rat is in a particular location: O'Keefe & Dostrovsky, *Brain Research* 34:171-175 (1971). https://doi.org/10.1016/0006-8993(71)90358-1
- Grid cells in the entorhinal cortex with firing fields forming a triangular (hexagonal) grid: Hafting et al., *Nature* 436:801-806 (2005). https://doi.org/10.1038/nature03721
- O'Keefe, M.-B. Moser and E. Moser shared the 2014 Nobel Prize in Physiology or Medicine for these discoveries. https://www.nobelprize.org/prizes/medicine/2014/summary/

Script 7: "Night Rinse"

Format: 50s short + song. **Topic:** neuroscience

Hook

"While you sleep, your brain does the washing up."

Shot list

1. Mat yawning, lights off; brain model on a nightstand starts to glow blue.
2. Card: 2013, mice studies show the brain's fluid spaces widen during sleep.
3. Animation: cerebrospinal fluid flows faster through the widened gaps, carrying waste out.
4. Card: waste includes beta-amyloid, a protein linked to Alzheimer's disease.
5. Card: 2019, human MRI shows slow pulses of fluid washing in during deep sleep.
6. Waves animation: slow brain waves, then blood pulls back, then fluid rushes in, in rhythm.
7. Mat under a blanket: "Sleep isn't lazy. It's cleaning day."

Lyrics: "Night Rinse"

Verse Lights go down, the gaps grow wide, Tide comes in from the other side, Sweeping out what the day left behind, A gentle rinse for a tired mind.

Chorus Night rinse, slow and deep, Waves roll in when I'm asleep, Wash it clean, start brand new, Dream on, brain, it's good for you.

Bridge Slow wave, blood ebbs, fluid flows, That's the rhythm when the eyelids close.

Science notes

- In sleeping (and anaesthetised) mice, interstitial space increased by ~60% and clearance of beta-amyloid increased: Xie et al., *Science* 342:373-377 (2013). https://doi.org/10.1126/science.1241224
- In humans during NREM sleep, slow neural waves are coupled to blood-volume changes and large pulses of CSF flow: Fultz et al., *Science* 366:628-631 (2019). https://doi.org/10.1126/science.aax5440
- Caution: the "brain washing" idea is strongly supported in mice; how much it drives waste clearance in humans is still being studied.

Script 8: "Water Bear"

Format: 50s short + song. **Topic:** animal kingdom

Hook

"This animal can be dried out like a raisin and come back to life."

Shot list

1. Microscope zoom into moss: a chubby eight-legged tardigrade waddles into view.
2. Size card: about half a millimetre long.
3. Drying animation: it curls into a "tun", nearly all its water gone.
4. Card: 2017, they make special floppy proteins (TDPs) that turn to glass and protect the cell.
5. Card: 2016, a tardigrade protein called Dsup protected human cells from X-ray damage in the lab.
6. Add water: the tun swells and walks off.
7. Mat holding moss: "Tiny, cute, and basically unbreakable."

Lyrics: "Water Bear"

Verse Eight little legs in a mossy bed, Dry spell comes, so I curl instead, Glassy proteins hold me tight, Wait for the rain, then I'm alright.

Chorus Water bear, water bear, Dry me out, I just don't care, Add a drop and here I go, Waddling on, slow-mo.

Bridge Dsup shield, X-rays fly, Tiny bear says "Not today, bye!"

Science notes

- Tardigrade-specific intrinsically disordered proteins (TDPs) are needed for desiccation tolerance and vitrify (form glass-like solids): Boothby et al., *Molecular Cell* 65:975-984 (2017). https://doi.org/10.1016/j.molcel.2017.02.018
- The Dsup protein from *Ramazzottius varieornatus* reduced X-ray-induced DNA damage in cultured human cells: Hashimoto et al., *Nature Communications* 7:12808 (2016). https://doi.org/10.1038/ncomms12808
- Tardigrades are typically up to about 0.5 mm long; the dried, curled state is called a "tun".

Script 9: "Spacetime Chirp"

Format: 50s short + song. **Topic:** astrophysics

Hook

"In 2015, the whole Earth stretched by less than a proton's width, and we heard it."

Shot list

1. Two black holes spiral together; a fabric grid ripples outward.
2. Card: 1.3 billion years ago, black holes of ~36 and ~29 Suns merged.
3. Card: ~3 Suns' worth of mass turned into gravitational-wave energy.
4. LIGO aerial shot: two L-shaped detectors, 4 km arms, in Washington and Louisiana.
5. Waveform plays as sound: a rising "whoop", the chirp.
6. Card: 14 September 2015, the first direct detection (GW150914), 100 years after Einstein predicted them.
7. Mat with headphones: "That's the sound of spacetime shaking."

Lyrics: "Spacetime Chirp"

Verse Two dark giants dancing near, Faster, faster, year by year, Then a crash no eye could see, Ripples racing out to me.

Chorus Spacetime chirp, whoop and gone, A billion years it travelled on, Stretch and squeeze, a tiny sign, Einstein's song, right on time.

Bridge Four-kilometre arms of light, Caught a whisper in the night.

Science notes

- First direct detection of gravitational waves, GW150914, 14 Sept 2015, from a binary black hole merger of ~36 and ~29 solar masses at ~410 Mpc (~1.3 billion years ago), radiating ~3 solar masses of energy: Abbott et al. (LIGO & Virgo), *Physical Review Letters* 116:061102 (2016). https://doi.org/10.1103/PhysRevLett.116.061102
- Peak strain ~1.0×10^-21; LIGO's arms are 4 km, so the change in length was far smaller than a proton: same paper.
- Einstein predicted gravitational waves in 1916.